

Riparian vegetation recovery in a 23 year chronosequence of clear-cuts along boreal headwater streams

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ABSTRACT

Recovery of understory vegetation after clearcut harvesting has received much attention but little is known about the recovery of riparian vegetation after forest harvesting along headwater streams, despite the critical role vegetation plays in the protection of biodiversity and water quality. We investigated the impacts of clearcut harvesting on riparian plant communities and their recovery along headwater streams in boreal mixedwood forests of northwestern Ontario, Canada. We studied 30 headwater streams (width 1–3 m), 24 in clearcut sites harvested 3–23 years previously and 6 streams in reference 90–100 year-old mature (unharvested) forests. Each stream was studied at two locations: (i) within clearcut areas and (ii) within the downstream unharvested locations (at the buffer zones) of the larger stream into which the headwater streams flow. We quantified species richness, abundance, diversity and evenness of riparian understory vegetation at each site. We used a nested model and Nonmetric Multidimensional Scaling (NMS) to examine the effect of years since clearcut (age) on understory plant community change. We assessed the number of original, newly recruited and locally eliminated species based on species presence/absence data from the sampled quadrats in clearcuts, downstream unharvested forests and reference forests. We found that clearcut harvesting had no immediate impact on overall species richness or diversity, but it was associated with changes in species composition. The post-harvest habitats were often dominated by ruderal invasive species. Some late-seral species were locally eliminated and showed no sign of recovery 23 years after harvesting. We conclude that although clearcutting along headwater streams does not affect the overall species diversity, it changes species composition by locally eliminating some understory plants, which in turn may cause significant change in the ecological services they provide. Hence, it is important to recognize the sensitivity of headwater systems during forest harvesting and subsequent vegetation recovery. Avoiding ground disturbance in head water systems during forest management, careful selective/partial harvesting in riparian areas and seeding of fast growing native plants immediately after clearcutting may help mitigate potential adverse effects of forest management in headwater systems.

1. Introduction

Favorable soil, water and available nutrients make riparian ecosystems highly productive and they are considered biodiversity hotspots (Naiman et al., 2005; Sabo et al., 2005), but they are subjected to a variety of natural and anthropogenic disturbances. In forest areas, clearcut harvesting is the most predominant anthropogenic disturbance and adds to the suite of natural disturbances influencing the structure and function of riparian areas (Naiman et al., 2000; Lamb et al., 2003; Mallik et al., 2013). Forest harvesting results in significant differences in plant communities compared to uncut forests beyond just the loss of

a portion of old-growth flora (Loya and Jules, 2008). It may result in vegetation typically dominated by ruderal species and in some instances, exotic species (Halpern et al., 1999). On the other hand, it may enhance natural regeneration of tree species, which prefer mineral substrates, facilitate recovery of shade tolerant herbs, planted trees and advanced regeneration (Roberts and Dong, 1993; McInnis and Roberts, 1994; Mallik et al., 2013, 2014).

The effects of forest harvesting on vegetation recovery following disturbance around headwater stream systems (first-order channels with catchments of < 100 ha, bankfull width 1–3 m and mean annual discharge < 57 l/s (Richardson and Danehy, 2007) have not been

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adequately studied even though headwater streams comprise almost 80% of total stream length in many drainage networks and drain a large portion of the catchment area (Naiman et al., 2005). Incomplete understanding of the benefits provided by the headwater streams, including biotic habitat, ecosystem services and water quality, often results in these features receiving limited attention from resource managers. Although headwater streams are small and numerous, they are closely connected to the terrestrial areas and are highly sensitive to environmental change and often not protected by unharvested riparian reserve or 'buffer' (MacDonald and Coe, 2007). It may be difficult and economically impractical to protect numerous headwater streams with riparian buffers in the same manner as larger streams. Although the volume of water flowing through an individual headwater stream is low, because of their ubiquitous presence and inter-connectedness in the landscape they are sensitive to forest management and environmental change and need special attention (Kahl, 1996).

Recovery of plant communities after forest harvesting is system specific and influenced by numerous biotic and abiotic factors (Battles et al., 2001; Sullivan et al., 2001; Costa and Magnusson, 2002; Roberts, 2004). Species richness and diversity has been found to increase following post-fire salvage logging (Greenberg et al., 1995). However, Duffy and Meier (1992) reported a lower post-harvest species richness and cover of herbaceous plants compared to primary forests even 87 years after harvesting. In most cases an initial decrease in diversity, particularly interior forest species, was followed by a quick recovery within a few decades (Meier et al., 1995; Hannerz and Hanell, 1997; Gilliam, 2002; Roberts and Zhu, 2002). However, some late-seral species may take decades or even centuries to recover from harvesting disturbance in managed forests (Meier et al., 1995; Goebel et al., 1999; Loya and Jules, 2008).

Disturbance intensity significantly influence the riparian plant colonization along boreal streams, with both species diversity and functional diversity reaching their peak under a moderate intensity of disturbance (Biswas and Mallik, 2010). However, findings of several authors suggest that the predicted peak in diversity at intermediate disturbance levels rarely occurs following anthropogenic disturbance (Mackey and Currie, 2001; Hughes et al., 2007; Fox, 2013) indicating that it may be contextual.

The diversity–disturbance relationship varies depending on the values of the disturbance aspects considered, e.g. frequency of disturbance over time, intensity of disturbance or the interactions between disturbance frequency and intensity (Hall et al., 2012). Forest harvesting in adjacent uplands does not strongly affect riparian vegetation composition along buffer-protected streams (Lamb, 2002). Changes in stream hydrology following disturbance in adjacent uplands influence the distribution and abundance of riparian species (Bendix, 1994; Naiman and Décamps, 1997), which significantly influences and structures riparian vegetation, instream biota and stream biogeochemical processes (King et al., 2009). Lateral connectivity of headwaters plays an important role in aquatic-terrestrial ecosystem functioning involving water and nutrient transport and riparian vegetation development (Hwang et al., 2012). Headwater streams (1–3 m wide) differ from mapped larger streams in terms of discharge rates, drought and flooding associated extremes, response to rainfall and flashiness (Baker et al., 2004). Since the hydrology of headwater streams, which are often not buffered and highly sensitive to soil disturbance from harvesting equipment (Mallik et al., 2011), differs from the larger streams protected by riparian buffers, the response of riparian understory plant communities along headwater streams to logging disturbance is expected to be different from that of larger streams. However, most riparian studies to date have been conducted on larger buffer protected streams.

In this study we examined the impacts of clearcut forest harvesting on the diversity and composition of riparian vegetation along unbuffered headwater systems with two main objectives: (a) to assess how clearcut harvesting alters the diversity and composition of riparian

vegetation and (b) to examine the pattern of vegetation recovery with time since harvesting. We hypothesized that (i) if clearcut harvesting alters riparian understory vegetation then floristic composition/diversity indices in younger clearcut sites will be significantly different from adjacent downstream unharvested sites and reference sites and (ii) ruderal species will dominate the recently harvested sites and be gradually replaced by old-growth species, with time since clearcut the compositional difference will decrease consequently older clearcuts and reference forests will have nearly similar floristic composition.

2. Methods

2.1. Study sites

Our study sites were within the watersheds of the Mackenzie, Current, Wolf and Kaministiquia rivers in northwestern Ontario, Canada (48°38'–48°50' N, 88°45'–89°23' W). The watersheds are part of the Thunder Bay plains eco-region, composed primarily of diabase, greywacke and shale bedrock formations with low, rolling relief overlain by thin glacial tills, exhibiting poor water retention and soil drainage (Wickware and Rubec, 1989). The area has a boreal temperate climate with mean temperature varying from –26 °C to –22 °C in January and 21 °C to 25 °C in July. Mean annual rainfall ranges between 700 mm and 850 mm (Baldwin et al., 2000).

Vegetation of the area consists of boreal mixedwood and conifer-dominated stands typical of the southern boreal forest (Rowe, 1972). Dominant tree species include white spruce (*Picea glauca*), balsam fir (*Abies balsamea*), white birch (*Betula papyrifera*), jack pine (*Pinus banksiana*) and trembling aspen (*Populus tremuloides*) with tamarak (*Larix laricina*), eastern white cedar (*Thuja occidentalis*) and black spruce (*Picea mariana*) occupying wetter sites (Rowe, 1972). The understory is dominated by large-leaved aster (*Aster macrophyllus*), bunchberry (*Cornus canadensis*) and blue bead lily (*Clintonia borealis*) (Stewart and Mallik, 2006). Based on the local ecosite classification guidelines the riparian vegetation along the studied streams can be broadly separated into three groups: *Calamagrostis canadensis* and *Carex aquatilis* dominated meadow marshes, *Alnus incana* dominated swamp thickets, and occasionally *Thuja occidentalis* dominated conifer swamps (Harris et al., 1996; Rankin, 2000).

2.2. Site selection

Potential study areas were first located using a GIS based flow accumulation model (ArcGIS, ArcMap version 9.2, ESRI Inc., Redlands, CA) derived from a 20 m resolution digital elevation model (Ontario Ministry of Natural Resources, 2005a). Subsequently, through extensive field visits, we selected 30 streams following a stratified random sampling method. Twenty-four of these streams were selected across a chronosequence of second-growth clearcuts with ages 3 to 23 years since harvesting. The ages of clearcuts were obtained from forest resource inventory data (Ontario Ministry of Natural Resources, 2005b). We selected six small streams located in 90–100-year-old mature forests with no adjacent forestry activities or fire as reference sites for comparison (Fig. 1).

Since small streams in the boreal forest are often characterized by missing channel features due to sub-surface flow (Hupp, 1986), we considered the following criteria in the final selection of study streams: (i) presence of a recognizable stream bed with upward slope on either side, (ii) bank full width up to 3 m, (iii) catchment area less than 100 ha, and (iv) direct connection with a larger stream. To minimize variation among sites we considered ecosite type, site class, landform, relief, drainage conditions, average slope and soil texture of the sampling sites.

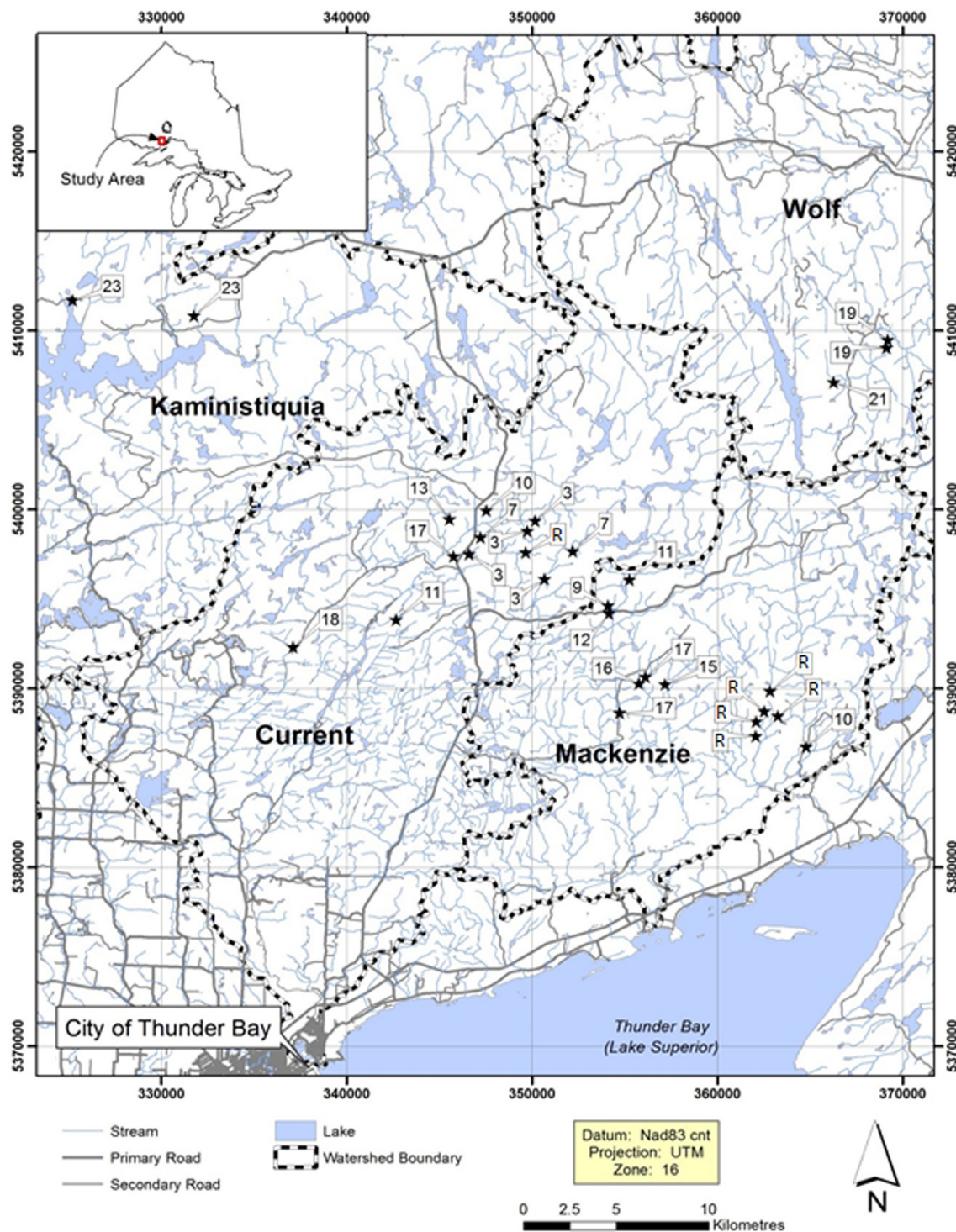


Fig. 1. Map of the study area in northeast of Thunder Bay, Ontario, Canada, showing study site locations of streams. Stars are stream locations, corresponding numbers indicate time (years) since harvesting and R stands for reference sites.

2.3. Sampling design and explanatory variables

Our study streams were not mapped on Ontario's 1:20,000 scale base maps and therefore did not have 30–90 m unharvested riparian reserves, or buffers, which are required on all mapped streams under Ontario's forest management guidelines (OMNR, 2010). However, all the sampled streams flowed into larger, mapped streams which did have a 30–90 m unharvested shoreline reserve allowing us to locate sampling sites in both clear-cut harvested and unharvested areas along

our small study streams (Fig. 2). We selected a 50 m section in the clear-cut area of each small stream at least 10 m upstream from the unharvested downstream end and divided it into ten 5 m segments from which 3 segments were selected at random. In each segment we established a transect perpendicular to the stream running through the riparian zone to the riparian-upland transition zone. The riparian-upland transition was identified based on topography (i.e. change from flatter to steeper slope) and a shift from wetland to upland plant communities (Harris et al. 1996). Transects were divided into

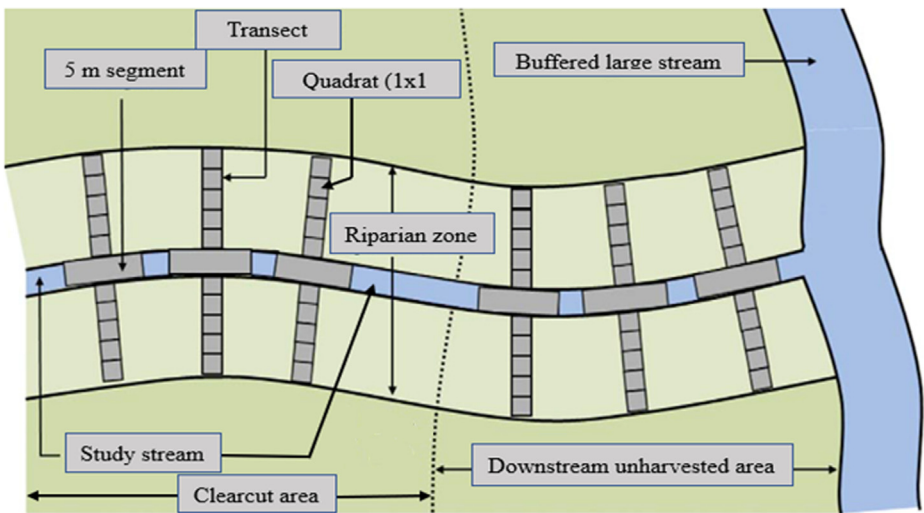


Fig. 2. A schematic diagram of the sampling design showing sampling protocol of the study at clearcut and downstream unharvested areas.

consecutive 1 × 1 m quadrats. Since the width of the riparian zones varied within and among streams, the number of quadrats per transect varied from 2 to 10. In the downstream unharvested section of each stream we established three additional transects: one at the mid-point between the large stream and clear-cut area and other two at 10 m up and down-stream of the center transect (see Fig. 2). This design allowed us to compare harvest to unharvested conditions within each stream to control for among stream variation in plant community structure. A similar sampling protocol was followed for small streams in unharvested, 90 to 100-year old mature forests, where stream segments were sampled at distances upstream from the point where the small stream joined a larger stream that were equivalent to the clear-cut streams. To avoid edge effects, we placed transects in the reference stream sites at least 50 m away from clear-cut (Murica, 1995; Dupuch and Fortin, 2013). To average the aspect effect on microclimate and species composition we established the sampling transects on both sides of the streams (Matlak, 1993; Chen et al., 1995; Burton, 2002).

We identified all vascular plants and mosses in each quadrat to species level and recorded their percent cover by visual estimation. For scientific names of vascular plants and mosses we followed Morton and Venn (1990) and Ireland et al. (1980) respectively. Unidentifiable plants in the field brought samples at laboratory and identified them comparing with the herbarium species at the Claud Garton Herbarium of Lakehead University except some grass, sedge and moss species that were grouped in to genus (see Appendix 3). We determined the number of original, newly recruited and locally absent species based on species presence/absence data from the sampled quadrats along the transects of clearcuts, downstream unharvested forests and reference forests. For each quadrat we also recorded the following physical and environmental variables: bank full width of stream (referred to as stream width), stream channel depth (referred to as stream depth), slope, aspect, distance of quadrat from stream, riparian width, exposed rock cover, coarse woody debris (CWD) cover (≥ 10 cm diameter), canopy exposure, ground exposure and depth of organic matter, which were used as explanatory variables in the analysis. We measured stream width at the point on the banks where the rooted, non-grass vegetation begins. Stream depth at bankfull was measured at the center of the stream using a meter stick. Riparian width was measured to the nearest 0.1 m at the point of the most abrupt shift from predominantly riparian vegetation to predominantly upland vegetation. At each quadrat exposed rock cover, CWD cover and ground exposure were determined by visual estimation. Canopy exposure for each quadrat was measured as the average of readings taken in each cardinal direction using a concave hemispherical Densiometer held at breast height (1.37 m). Depth of

organic matter at each quadrat was measured from 30 × 30 × 30 cm deep soil pits.

2.4. Statistical analysis

We calculated simple univariate measures at the stream scale to determine the level of alpha-diversity to allow for comparison between reference and clearcut sites of different ages since clearcut. We determined three diversity indices: species richness (*S*), Shannon-Wiener index of diversity (*H'*), and evenness (*E*), as well as percent cover (abundance) for all species and within major life-form groups: trees, shrubs, herbs, ferns, grass and moss. To evaluate the impacts of clearcut harvesting on diversity indices and abundance over time since harvesting, we used a linear regression model with time since harvesting as the independent variable and the difference in diversity indices and abundance between upstream clearcut and associated downstream unharvested locations within each stream as the dependent variable. This analysis excluded reference sites because we did not have exact ages for reference areas. Because the regression analyses showed no consistent or significant relationships between time since harvest and any index of diversity or abundance, except for grasses (Table 1), in the subsequent analyses we used categorized sites of 3 age classes: early

Table 1
Results of regression analyses with differences in life-form-based diversity indices between clearcut and corresponding downstream unharvested locations as the dependent variable and time since harvesting as the predictive variable. Values in bold indicate significant relationships at 95%.

Lifeform		Diversity	Evenness	Richness	Abundance
All species	R square	0.156	0.049	0.095	0.050
	Sig.	0.056	0.299	0.143	0.295
Tree	R square	0.109	0.114	0.015	0.075
	Sig.	0.115	0.106	0.571	0.197
Shrub	R square	0.072	0.001	0.154	0.091
	Sig.	0.204	0.915	0.058	0.151
Herb	R square	0.021	0.048	0.045	0.140
	Sig.	0.503	0.303	0.317	0.071
Fern	R square	0.011	0.033	0.044	0.054
	Sig.	0.630	0.397	0.327	0.272
Grass	R square	0.387	0.297	0.369	0.300
	Sig.	0.001	0.006	0.002	0.006
Moss	R square	0.013	0.009	0.036	0.063
	Sig.	0.602	0.658	0.373	0.236

clearcut sites (ages 3–7 years, $n = 6$) as age class 1, late clearcut sites (ages 18–23 years, $n = 6$) as age class 2 and reference sites (age 90–100 years, $n = 6$) as age class 3 which allowed us to evaluate plant community changes over time since harvest in relation to a reference condition. To assess the vegetation recovery over time since harvesting, we used a nested model of ANOVA with type III sum of squares. We used age classes and disturbance types (upstream clearcut and downstream unharvested forest) as fixed factors, streams nested within age classes as random factor and species richness, diversity, evenness and abundance as response variables. Prior to statistical analysis, the explanatory variables measured at each quadrat were averaged to produce stream-level estimates. The data residuals were checked for normality by Kolmogorov-Smirnov test and homogeneity of the variances by Levene test. All analyses were performed using IBM SPSS version 25.0 (SPSS, 2017).

We explored the overall structure and trends in the species compositional data using Non-Metric Multidimensional Scaling (NMS), using the autopilot option with a slow and thorough analysis and the default settings. The hypothesis of no significant floristic differences between the age classes was tested using Multiple Response Permutation Procedure (MRPP) (Zimmerman et al., 1985). We used an Indicator Species Analysis (INSPAN) following the method of Dufrêne and Legendre (1997) to identify the affinity of a species for a particular age class and disturbance type based on a comparison of the mean relative frequency and cover of a species in a pre-defined group to the same values calculated for all groups. For each individual species the strength of its association with a specified age group was tested with Monte Carlo permutations of 4999 runs where samples are randomly reassigned to groups and the indicator values are recalculated. Mean percent cover for all understory species were used as the response variable. Species found in less than 5% quadrats were removed from the analysis. A total of 126 species was used in the analysis. All multivariate tests were carried out using the PC-ORD program version 6 (McCune and Mefford, 2011).

3. Results

We recorded a total of 259 plant species belonging to 146 genera and 71 families. The maximum number of species was recorded from the family Cyperaceae (25), followed by Rosaceae (21) and Compositae (20). The number of species was highest for herbs (85) followed by mosses (55), shrubs (46) and grasses/sedges (44). *Alnus incana*, *Acer spicatum*, *Rubus idaeus*, *Corylus cornuta* and *Cornus stolonifera* were the most abundant shrubs in the area. Among the herbs *Aster macrophyllus*, *Clintonia borealis*, *Thalictrum dasycarpum*, *Galium asprellum* and *Mertensia paniculata* were the most abundant. *Calamagrostis canadensis* and *Carex intumescens* were the most dominant grass/sedge species. *Athyrium filix-femina* and *Eurhynchium pulchellum* was the most abundant fern and moss, respectively. *Carex crinita* was fairly common in the area although it is considered locally rare in the Thunder Bay District (Thunder Bay Field Naturalists, 1998). *Rubus idaeus* and *Aster macrophyllus*, two upland species, were found to be dominant in the clearcut sites immediately after harvesting.

3.1. Species richness, abundance, diversity and evenness

The results of regression analyses showed that differences in diversity indices between upstream clearcuts and downstream unharvested forests had no significant relationship with time since harvesting except for the grasses (Table 1 and Fig. 3). In early clearcuts grass diversity indices were very high compared to corresponding downstream unharvested forests but the differences gradually declined with time since harvesting.

We found no differences in overall species richness within the age classes, either in clearcuts or in downstream unharvested forests (Table 2). However, richness of shrubs was significantly higher in

downstream unharvested forests of age class 1 and clearcuts of age class 2 compared to clearcuts of age class 1 (Fig. 4). Richness of herbs was lower in age class 2 compared to age class 1 and reference forests. Richness of grasses was higher in clearcuts compared to downstream unharvested forests and reference forests. The highest richness of grasses was observed in clearcuts of age class 1 and the lowest in the downstream location of reference forests. Richness of mosses was significantly higher in age class 2 than in age class 1, which varied significantly between disturbance types, high in downstream unharvested forests and low in clearcuts (Table 2 and Fig. 4).

There was no significant difference in overall species abundance or that of trees, shrubs, herbs and ferns (Table 2). However, the abundance of grasses and mosses were significantly related to both age class and disturbance type. Abundance of grass was significantly lower in reference forests and in downstream unharvested forests. Grass abundance was lowest at the upstream locations of reference forests and highest at early clearcuts. Abundance of moss was significantly lower in age class 1. However, in age class 2 it was similar to the reference forests. Compared to downstream unharvested forests, moss abundance was lower in the upstream clearcuts (Table 2 and Fig. 4).

Diversity of herbs, grasses and mosses varied among the age classes. In age class 2, herbs had lowest diversity while mosses had the highest diversity. Grass diversity was very low in the reference forests. Diversity of tree seedlings was lowest in downstream unharvested forests of age class 1 and highest in clearcuts of age class 2. Grasses had higher diversity in the clearcuts compared to downstream unharvested forests, which was opposite for the mosses. Highest diversity of grasses was in clearcuts of age class 1 (Table 2 and Fig. 4).

Evenness of herbs, grasses and mosses was significantly related to age class (Table 2). Evenness of herbs was highest in reference forests while grasses were least evenly distributed in reference forests. Mosses were most evenly distributed in age class 2. Tree seedlings were most evenly distributed in the clearcuts of age class 2 while grasses were most evenly distributed in clearcuts of age class 1 and downstream unharvested forests of age class 2. Evenness of grasses was significantly higher in clearcuts than downstream unharvested forests (Table 2 and Fig. 4).

3.2. NMS ordination, MRPP and Indicator Species Analysis

Plant species composition for all stream sites was summarized in 3 dimensions using Nonmetric Multidimensional Scaling (NMS). The final stress was 16.09442, which is fairly low and correlations (r^2) between ordination distances in the final solution and distances in the original n -dimensional space were 0.449, 0.224 and 0.124 for the first, second and third axes, respectively, resulting in a cumulative r^2 of 0.797, accounting for most of the variance structuring the data set. Since axis 3 explained only 12% of the total variance, here we interpret only axes 1 and 2. In the ordination space, axis 1 is strongly negatively correlated with canopy exposure and positively correlated with ground exposure. Axis 2 is positively correlated with distance from stream, riparian width and depth of organic matter (Fig. 5). Species with high scores on NMS axis 1 appear in the right quadrant of the ordination (Fig. 6). These are mostly shade tolerant species e.g., *Acer spicatum*, *Circaea alpina*, *Osunda claytoniana*, *Viola nephrophylla*, etc. Species with low scores on NMS axis 1 (Fig. 6) appear in the left quadrant of the ordination. These plants attained their maximum abundance in exposed sites following clearcut harvesting, e.g. *Epilobium angustifolium*, *Calamagrostis canadensis*, *Equisetum pratense*, *Carex lasiocarpum*, *Aster ciliolatus*, etc. Plants with low scores on NMS axis 2 are mostly typical of disturbed sites like *Cinna latifolia*, *Oryzopsis asperifolia*, etc. *Polytrichum commune* grew well on exposed rocks. Plants preferring moist upland sites appeared towards the high score end of NMS axis 2 (Fig. 6).

The difference in floristic composition between sites of different age classes was significant based on MRPP analysis ($T = -25.758677$, $p < 0.001$) and showed low within-group homogeneity ($A = 0.1069$).

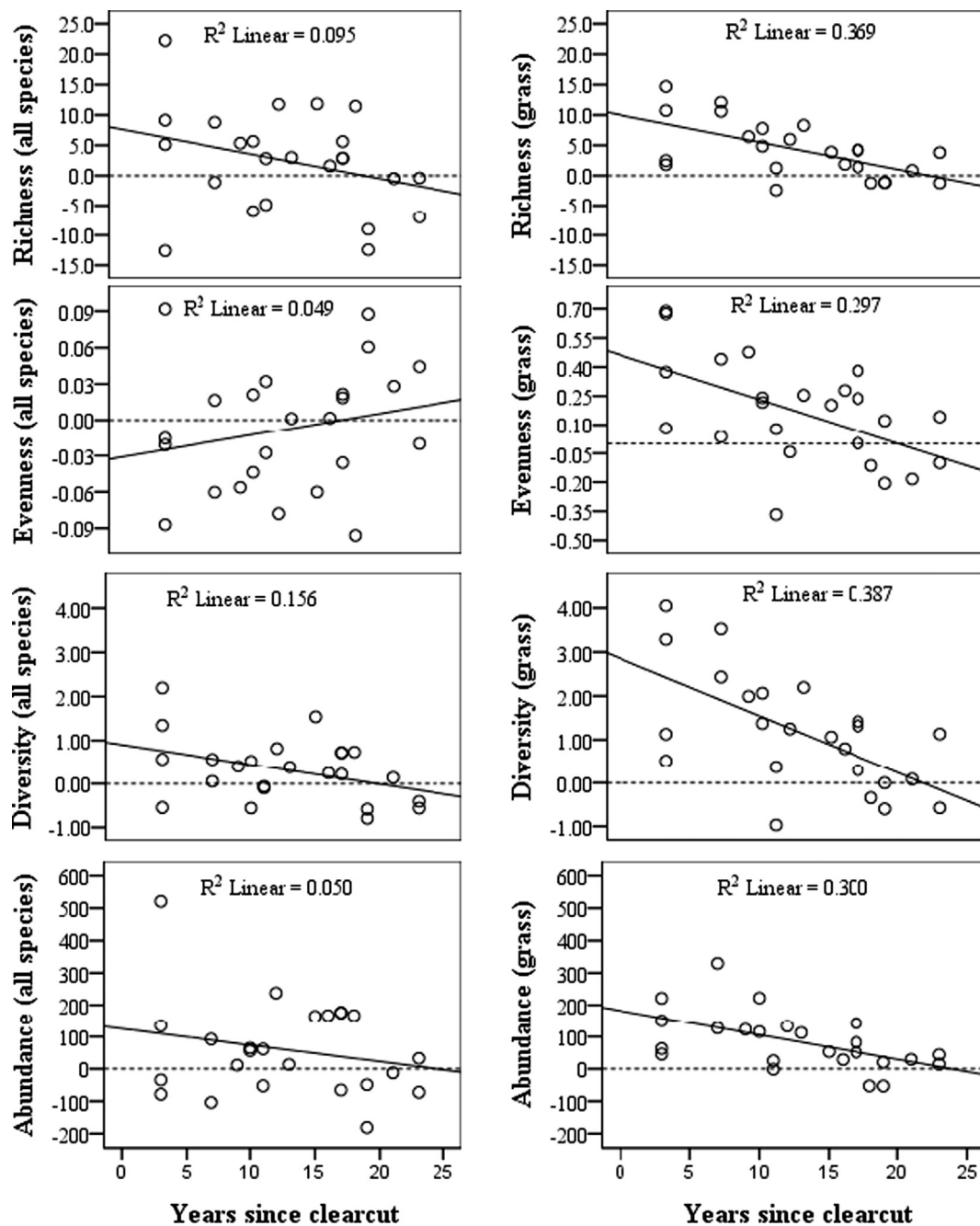


Fig. 3. Relationship between harvesting induced changes in diversity indices and time since harvesting. In the regression analyses differences in diversity indices between clearcut and corresponding downstream unharvested sites were used as the response variable and time since harvesting (3–23 years) as the predictor variable.

This indicates an overall difference in riparian community composition at different stages of recovery from harvesting; however, there was also a great deal of variation within age classes.

Indicator Species Analysis showed that out of 126 species used in the analysis, 21 were significantly associated with a particular age class. Twenty-five species showed almost a complete lack of association with any particular age class. Most of the species associated with the young clearcut sites were ruderal shade intolerant species. Mosses showed special affinity for downstream unharvested sites. Commonly known as mid-seral species, *Lycopodium dendroideum* was associated with the downstream unharvested forests of age class 1. *Rhizomnium pseudopunctatum* was the only species significantly associated with the reference forest at the upstream location (Table 3). Likewise, *Acer spicatum* had the highest frequency and abundance in reference forests.

3.3. Changes in species composition following harvesting

Harvesting disturbance was associated with an increase in total number of species, both at clearcut and downstream unharvested sites compared to the reference forests (Appendix 1). However, the magnitude of increase was highest in clearcuts of age 3–7 years followed by age 18–23 years. The highest number of newly recruited species (74) was recorded in the clearcuts of age 3–7 years, mostly herbs (34) and grasses (16). The species that disappeared from the early disturbed sites in both clearcuts and downstream unharvested forests were mostly mosses. The number of newly recruited species was higher than the number of species that disappeared irrespective of time since disturbance or locations (Appendix 1). In this study, species such as *Betula papyrifera*, *Amelanchier* spp., *Diervilla lonicera*, *Salix petiolaris*, *Vaccinium*

Table 2

Results of nested ANOVAs for the effects of time since harvesting, disturbance types and their interactions on species richness, abundance, diversity, and evenness in the boreal riparian forests of Ontario. In the analyses, diversity indices were used as dependent variables, age classes (3–7 years, 18–23 years, and reference forests) and disturbance types (upstream clearcut and downstream unharvested) as fixed factors, and streams nested within age classes as random factor. Values in bold are significant at 0.05.

Life form	Sources of variation	df	Species richness		Abundance		Diversity		Evenness	
			F value	p value	F value	p value	F value	p value	F value	p value
All species	Age class	2	0.704	0.510	0.458	0.641	1.734	0.210	0.024	0.977
	Stream (Age class)	15	1.744	0.146	2.155	0.074	2.058	0.087	1.518	0.214
	Disturbance type	1	0.431	0.522	0.304	0.590	0.923	0.352	0.008	0.932
	Age class × Disturbance type	2	1.242	0.317	0.823	0.458	2.764	0.095	0.512	0.609
Tree	Age class	2	0.266	0.769	0.448	0.645	1.894	0.185	2.191	0.146
	Stream (Age class)	15	1.901	0.180	1.123	0.449	1.493	0.223	1.447	0.241
	Disturbance type	1	0.007	0.937	0.186	0.673	0.424	0.525	0.317	0.582
	Age class × Disturbance type	2	0.030	0.971	0.859	0.446	4.886	0.023	4.999	0.022
Shrub	Age class	2	0.139	0.871	2.243	0.140	0.734	0.497	2.273	0.137
	Stream (Age class)	15	3.822	0.007	0.666	0.780	3.616	0.009	2.381	0.052
	Disturbance type	1	0.011	0.918	0.141	0.713	0.013	0.912	0.005	0.946
	Age class × Disturbance type	2	4.548	0.029	2.936	0.084	1.954	0.176	0.061	0.941
Herb	Age class	2	7.705	0.005	2.743	0.097	9.423	0.002	4.250	0.034
	Stream (Age class)	15	1.678	0.163	2.677	0.033	1.379	0.271	1.463	0.235
	Disturbance type	1	0.525	0.480	1.706	0.211	0.443	0.516	0.198	0.663
	Age class × Disturbance type	2	1.075	0.366	3.255	0.067	0.700	0.512	0.449	0.647
Fern	Age class	2	1.367	0.285	3.567	0.054	0.582	0.569	0.905	0.423
	Stream (Age class)	15	1.497	0.222	2.277	0.061	1.338	0.293	1.430	0.252
	Disturbance type	1	1.420	0.252	4.349	0.054	2.898	0.108	2.535	0.131
	Age class × Disturbance type	2	1.922	0.181	3.391	0.060	2.362	0.127	1.761	0.204
Grass	Age class	2	6.155	0.011	8.352	0.004	5.612	0.015	3.811	0.046
	Stream (Age class)	15	3.079	0.018	2.471	0.045	2.627	0.035	3.017	0.020
	Disturbance type	1	19.435	0.001	18.682	0.001	18.095	0.001	8.747	0.010
	Age class × Disturbance type	2	11.023	0.001	16.520	< 0.001	13.292	< 0.001	6.799	0.008
Moss	Age class	2	4.752	0.025	4.761	0.025	6.420	0.010	5.135	0.020
	Stream (Age class)	15	1.378	0.271	2.607	0.037	1.137	0.403	0.877	0.599
	Disturbance type	1	8.153	0.012	10.899	0.005	7.762	0.014	3.431	0.084
	Age class × Disturbance type	2	0.664	0.529	0.651	0.536	0.306	0.741	0.542	0.593

angustifolium, *Cirsium muticum*, *Epilobium angustifolium*, *Hieracium aurantiacum*, *Carex crinita*, *C. houghtoniana* and *Calamagrostis canadensis* were found to invade the clearcut sites whereas several old-growth species like *Actaea rubra*, *Atrichum* spp., *Dicranum montanum*, *Gentiana rubricaulis*, *Distichium capillaceum* and *Myurella julacea* locally disappeared after clearcut harvesting.

We also observed differences in life form-based dominance following harvesting (Appendix 2). In clearcut sites mean total grass cover was very high (26%) at the age 3–7 years and then declined to 13% at 18–23 years, compared to 6% in the reference forests. Herb cover was high in the 3–7-year age class and became lower than that of the reference sites in the 18–23 year age class. Moss and shrub cover initially declined (8% and 26% compared to 15% and 39% in reference sites, respectively) but fully recovered by 18–23 years (14% and 40%, respectively). Fern cover was reduced to 3% at 3–7 years then increased to 9% at age 18–23 years. Variation in life form-based dominance was comparatively lower in the downstream unharvested forests than clearcuts (Appendix 2). A shift in species dominance was observed with time since harvesting. *Rubus idaeus* was the most dominant species in clearcuts of age 3–7 years, while *Alnus incana* became dominant in clearcuts at 18–23 years and in downstream unharvested forests associated with age classes 3–7 and 18–23 years. *Acer spicatum* was the most dominant species in the reference forests of both upstream and downstream locations (Appendix 3).

4. Discussion

We found that time since clearcut harvesting along headwater streams had no significant effect on overall diversity of riparian understory vegetation. Although statistically not significant, the overall

species richness and diversity were higher immediately after harvesting (3–7 years) compared to the older age class (18–23 years) and in the reference forests. This short-term increase in species richness and diversity might be due to the survival of disturbance tolerant generalist species along with the addition of newly invading species immediately after harvesting, since moderate to high severity disturbance results in higher richness of exotic invasive species and native ruderal species (Crawford et al., 2001). A similar conclusion was drawn from another study in the Acadian forest region at southeastern New Brunswick by Roberts and Zhu (2002) reporting 250% increase in the number and abundance of disturbance-adapted species at high severity disturbance sites (clearcutting followed by mechanical site preparation and planting) compared to 200% increase in moderately disturbed sites (clearcutting followed by natural regeneration). Clearcut harvesting in our study also caused local elimination of some late-seral species such as *Actaea rubra*, *Atrichum* spp., *Dicranum montanum*, *Gentiana rubricaulis*, *Distichium capillaceum* and *Myurella julacea*. The absence of these species from all post-harvest age classes provides evidence of a lack of recovery over 23 years after harvesting. Although clearcutting had no immediate impact on overall alpha richness or diversity, richness and diversity of residual plants declined after canopy removal, which does not recover even over 54 years of secondary succession (Moola and Vasseur, 2004). In our study we found some species that disappeared immediately after clearcutting recovered in 7 years but some others took 23 years to re-establish. Some ruderal species invaded the area after harvesting and were present 23 years after harvesting while some others disappeared at different stages of habitat recovery (Appendix 2), which might lead to a compositional change. These results are consistent with the findings of Peterken (1996) and Battles et al. (2001) who showed that although overall species richness and

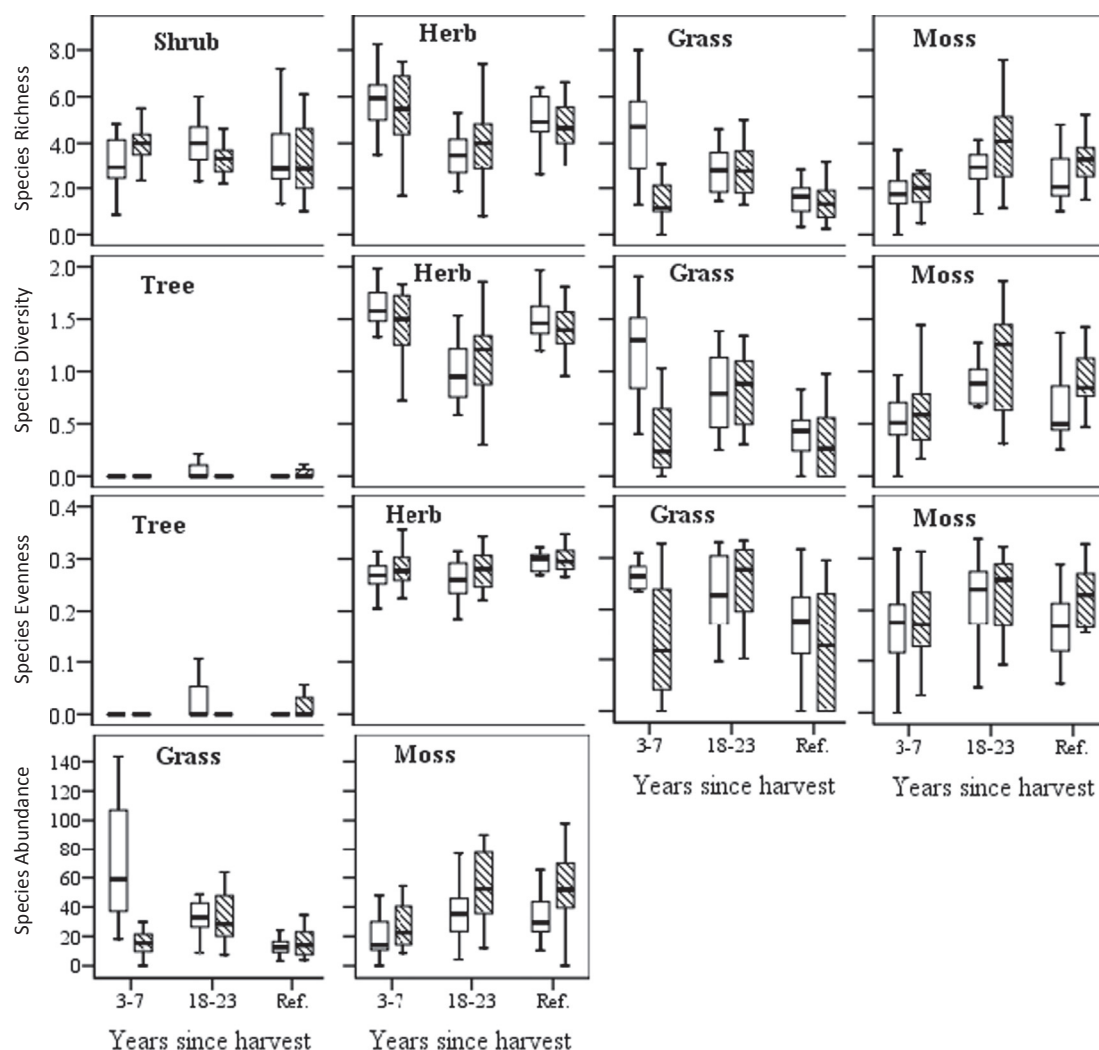


Fig. 4. Species richness, abundance, diversity and evenness at early clearcut (3–7 years since harvesting), late clearcut (18–23 years since harvesting) and reference (no evidence of harvesting disturbance in last 90–100 years) sites. The shaded boxes represent downstream unharvested locations while unshaded boxes represent clearcut/upstream locations. Horizontal bars are medians and boxes are quartiles (25–75).

diversity recovers rapidly during secondary succession, many of the typical residual species get eliminated from the plant communities in secondary forests. Through the loss of a portion of old-growth flora immediately after logging, clearcut harvesting results in compositional differences between immediate and late-seral stands, which persist for many decades after clearcutting (Loya and Jules, 2008; Moola and Vasseur, 2004). However, pre-disturbance species richness and evenness and bryophyte dominance influence the stability of the understory vegetation composition following harvesting (MacDonald et al., 2015). In one study Mallik et al. (2013) found no significant effect of partial harvesting along riparian buffers on species richness and diversity. However, in a subsequent study they found a denser and diverse woody plant community in partially harvested riparian buffers compared to unharvested buffers or reference forests (Mallik et al., 2014). Although a significant number of residual species is lost after clearcutting, diversity indices does not change due to the recruitment of an equal or greater number of newly invading species (McLachlan and Bazely, 2001), which is in line with our findings.

In this study we observed no significant difference in overall species abundance and species diversity between clearcuts and downstream unharvested forests, except higher diversity of grass in clearcuts. These results contradict with the findings of Biswas and Mallik (2010) who reported higher species diversity and functional diversity of riparian understory vegetation in buffer sites (equivalent to our downstream

unharvested forests) compared to 3 to 6-year-old clearcuts and reference forests. We found many invading ruderal species added to the community along with the disturbance-tolerant generalist species in younger (3–7 years) clearcuts, resulting no overall change in species diversity despite the elimination of disturbance intolerant species. Riparian vegetation is resilient to changes in microclimate following removal of the adjacent forest canopy (Brosfokske et al., 1997). They have mechanisms to survive flooding due to their tolerance for anoxic rooting zones (Blom and Voesenek, 1996). Many of these species also possess efficient dispersal, colonization and establishment strategies such as hydrochory and ability to rapidly colonize bare grounds by seed and clonal regeneration allowing rapid recovery after disturbance (Naiman and Décamps, 1997). These strategies help increase species richness in clearcuts at early successional stages compared to late successional stage and in downstream unharvested forests. In this context one must consider the effect of forest harvesting on local elimination and recovery of plant functional types (life-forms) related to ecosystem services.

Our MRPP results show that age class 1 of the downstream unharvested forests and age class 2 of the clearcuts differed the most from other combinations of age classes and disturbance types. This may be attributed to local elimination of some shade tolerant species from the downstream unharvested forests of age class 1 due to canopy opening caused by microclimatic edge effect (Stewart and Mallik, 2006;

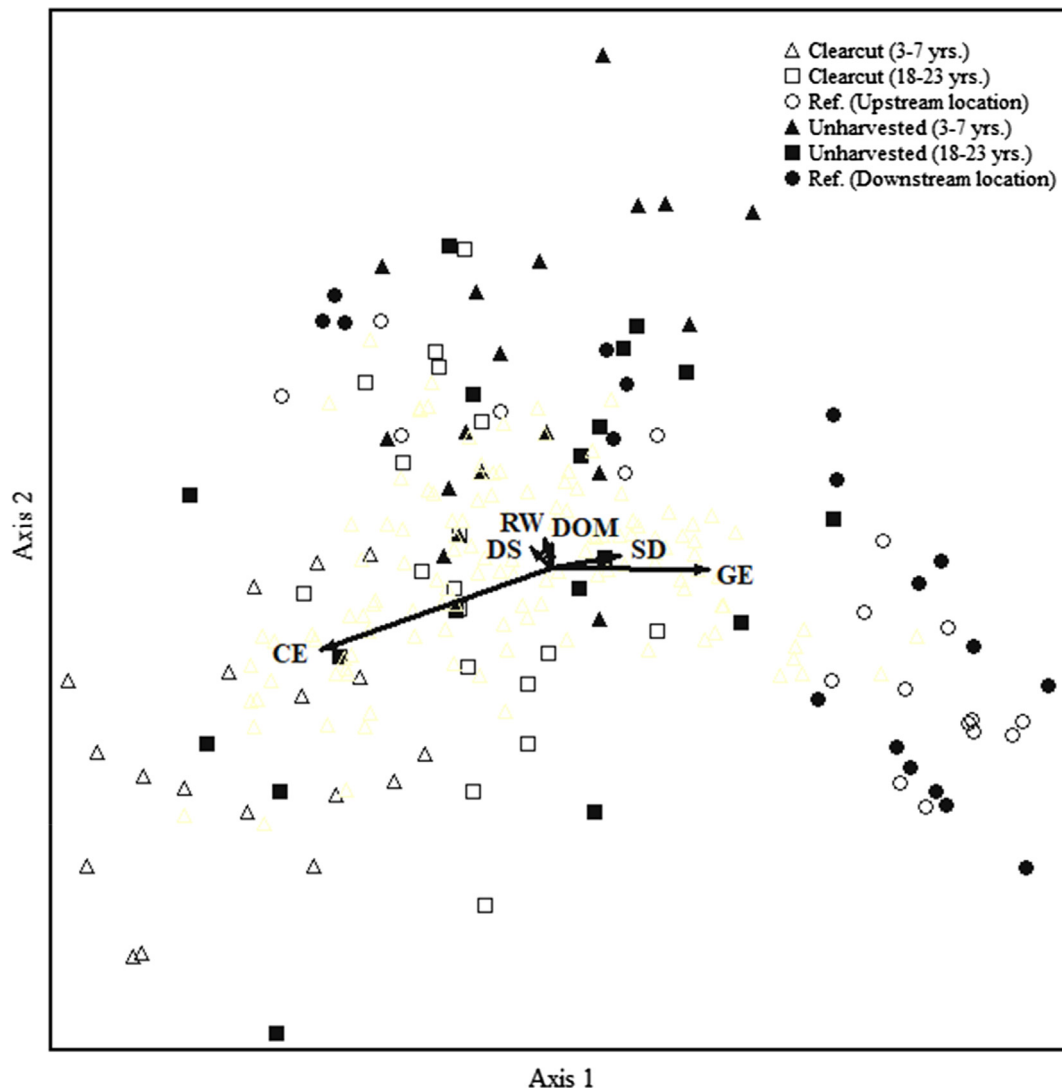


Fig. 5. NMS ordination of average species composition within age classes and disturbance types. Younger sites (3–7 years) are structured towards high canopy exposure and older sites (18–23 years) along with downstream unharvested and reference sites towards low canopy exposure and high ground exposure in the ordination space. Ref. stands for reference sites. CE = canopy exposure, GE = ground exposure, SD = stream depth, DOM = depth of organic matter, RW = riparian width and DS = distance from stream.

Braithwaite and Mallik, 2011). The edge effects in the form of significant difference in understory moss (*Hylocomium splendens* and *Polytrichum commune*) growth, ground temperature, relative humidity, vapor pressure deficit and light condition between clearcuts and adjacent uncut buffers was found to penetrate 20–30 m into the buffer. Similarly depth of edge effect for vascular and non-vascular plants can extend up to 30 m from clearcut edge into riparian buffer (Braithwaite and Mallik, 2011).

In our clearcut sites of age class 2 some of the initially invaded ruderal species might have disappeared due to decreased canopy exposure at this stage and failure of some late-seral species to re-establish. Immediately after harvesting herb and grass cover was very high, which declined gradually with time. Most of these species are shade intolerant (Humbert et al., 2007). On the other hand, cover of shade tolerant ferns and mosses was initially low in clearcuts but increased gradually with time. Changes in microclimate following canopy removal may exceed the tolerance of many non-vascular plants (Fenton and Frego, 2005; Hylander, 2005; Åström et al., 2007), causing decline in bryophyte cover (Fenton and Frego, 2005; Nelson and Halpern, 2005; Hart and Chen, 2008). Invading species are predominantly intolerant to shade and rapid reduction of light availability in post-clearcut successional

communities results in their elimination (Klinka et al., 1985). Relatively open canopy of immature stands favors the persistence of invading ruderal species with moderate shade tolerance (De Grandpré et al., 2000). Richness of shrubs and herbs attain their peak at 2–5 years after harvesting and then starts declining (Schoonmaker and McKee, 1988).

We found *Carex lasiocarpum*, *Polytrichum commune*, *Carex canescens*, *Glyceria borealis*, *Prunus pensylvanica*, *Aster nemoralis*, *Carex laxiflora*, *Elymus trachycaulus* and *Prunus virginiana* to be associated with very young (3–7 years) clearcuts, most of which are early successional grass/sedge species (Legasy et al., 1995; Beaudry et al., 1999). *Aster nemoralis* grows in areas of anthropogenic disturbance with wet soils exposed to sun. Shrub species such as *Prunus pensylvanica* and *P. virginiana*, associated with the early recovery stage, typically grows along the clearings of disturbed sites (Legasy et al., 1995; Beaudry et al., 1999). *Polytrichum commune* predominantly prefers mineral soils and is commonly found in sunny areas with little to no shade and thus frequent in exposed sites following harvesting (Legasy et al., 1995; Beaudry et al., 1999). No species was found to be associated with the late (18–23 years) recovery sites, which might be an indication of a transitional period. Species with special affinity to downstream unharvested forests were mostly mosses, growing well in partially shaded moist to dry locations along forest

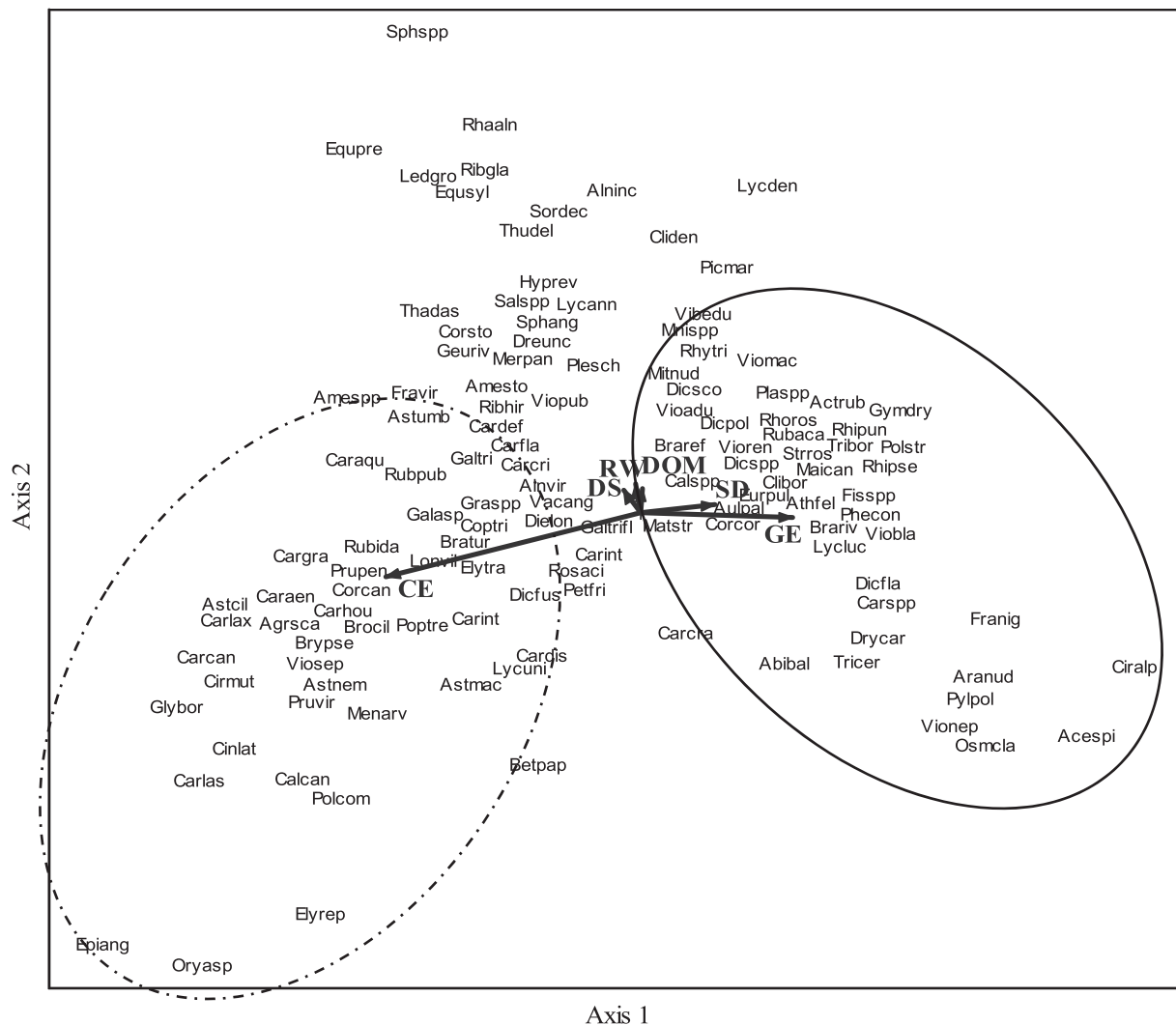


Fig. 6. NMS species ordination using the mean percent cover of 126 species within 18 streams. Species scores are shown for the first two axes of the ordination. Axis 1 explains 44.9% and Axis 2 explains 22.4% variation of the species data. Species are represented by codes with first three letters of the generic name followed by first three letters of the species name. Species within the dashed and compact ovals are mostly shade intolerant ruderal and shade tolerant old-growth species, respectively (Humbert et al., 2007).

edges and openings. Richness of mosses declines following clearcut harvesting (Baldwin et al., 2012). Perennial herbs *Trientalis borealis* and *Viola macloskeyi* were also associated with the downstream unharvested forests, which prefer shaded moist woods, bogs and fens (Legasy et al., 1995; Beaudry et al., 1999). Mosses such as *Rhizomnium pseudopunctatum* and *Brachythecium rivulare* had special affinity to undisturbed reference sites of upstream and downstream locations, respectively. Perennial herb *Circaea alpina*, a shade tolerant species growing well in mossy bogs was associated with the reference forests in the downstream location (Legasy et al., 1995; Beaudry et al., 1999). We observed a shift in species dominance with time since harvesting. After harvesting *Rubus idaeus* became the dominant species, which is an early successional species. But at age 18–23 years *Alnus incana*, once established maintains dominance for a long time by stem-base re-sprouting (Bell, 1991; Huenneke, 1987), became the dominant species replacing *Rubus idaeus*. *Acer spicatum* was the most dominant species in the reference forests of both upstream and downstream locations.

5. Conclusions and management implications

We showed that immediately after clearcut harvesting riparian vegetation becomes dominated by shade-intolerant ruderal species, which

is in agreement with our prediction that ruderal species will be dominating the recently harvested sites. Although some of these species locally disappeared, others persisted for a long time. We found no evidence of recovery of some typical riparian species 23 years since clearcut. Although clearcut harvesting along small headwater stream does not seem to affect the overall diversity of riparian vegetation, it alters species composition and dominance. Following harvesting a shift in species dominance may cause long-term effects on the structure and composition of riparian vegetation, which in turn may cause significant changes in the ecological services they provide. Therefore, it is important to consider the habitat sensitivity of headwater streams and associated riparian vegetation response to forest management action plans in order to safeguard the biotic and ecosystem integrity. The results of this study clearly illustrate the importance of considering small headwater streams and their riparian zones during forest management planning and operations. In addition to the importance of maintaining riparian areas to prevent impacts to streams, such as erosion and sedimentation, disturbance in these areas may result in a reduction in native biodiversity of certain lifeform/functional group which can persist for many years. Management practices that minimize riparian disturbance, such as using riparian buffers or careful operations to minimize disturbance and maintain understory plant communities, can

Table 3

Affinity of species for a particular age class and disturbance type as revealed by the Indicator Species Analysis. For each species the significance of the association was tested in a Monte Carlo permutation test (4999 runs) following the method of [Dufrene and Legendre \(1997\)](#). Species found in less than 5% quadrats were removed from the analysis.

Age class	Upstream (Clearcut location)			Downstream (Unharvest location)		
	Species	IV	p value	Species	IV	p value
Early (3–7 years)	<i>Carex lasiocarpum</i>	47.4	0.0002	<i>Climacium dendroides</i>	29.1	0.0012
	<i>Polytrichum commune</i>	23.9	0.0010	<i>Lycopodium dendroideum</i>	21.7	0.0030
	<i>Carex canescens</i>	22.9	0.0016	<i>Amelanchier stolonifera</i>	21.5	0.0092
	<i>Glyceria borealis</i>	27.4	0.0024	<i>Trientalis borealis</i>	23.2	0.0380
	<i>Prunus pensylvanica</i>	23.3	0.0044			
	<i>Aster nemoralis</i>	22.6	0.0064			
	<i>Carex laxiflora</i>	16.3	0.0360			
	<i>Elymus trachycaulus</i>	14.5	0.0440			
Late (18–23 years)	<i>Prunus virginiana</i>	12.7	0.0450			
				<i>Viola macloskeyi</i>	31.3	0.0014
				<i>Dicranum polysetum</i>	24.2	0.0030
				<i>Dicranum scoparium</i>	21.8	0.0072
				<i>Hypnum revolutum</i>	18.0	0.0324
Reference				<i>Mnium</i> spp.	18.2	0.0456
	<i>Rhizomnium pseudopunctatum</i>	23.2	0.0240	<i>Brachythecium rivulare</i>	26.9	0.0044
				<i>Circaea alpina</i>	21.3	0.0058

Note: IV is the indicator value, proportional abundance of a species in a particular group relative to the abundance of the species in all groups.

serve to mitigate these risks. Additional research is needed to better understand how plant community composition recovers from natural disturbances such as fire and how these impacts compare to harvesting.

Conflict of interest

The authors declare no conflict of interest, financial or otherwise.

Acknowledgements

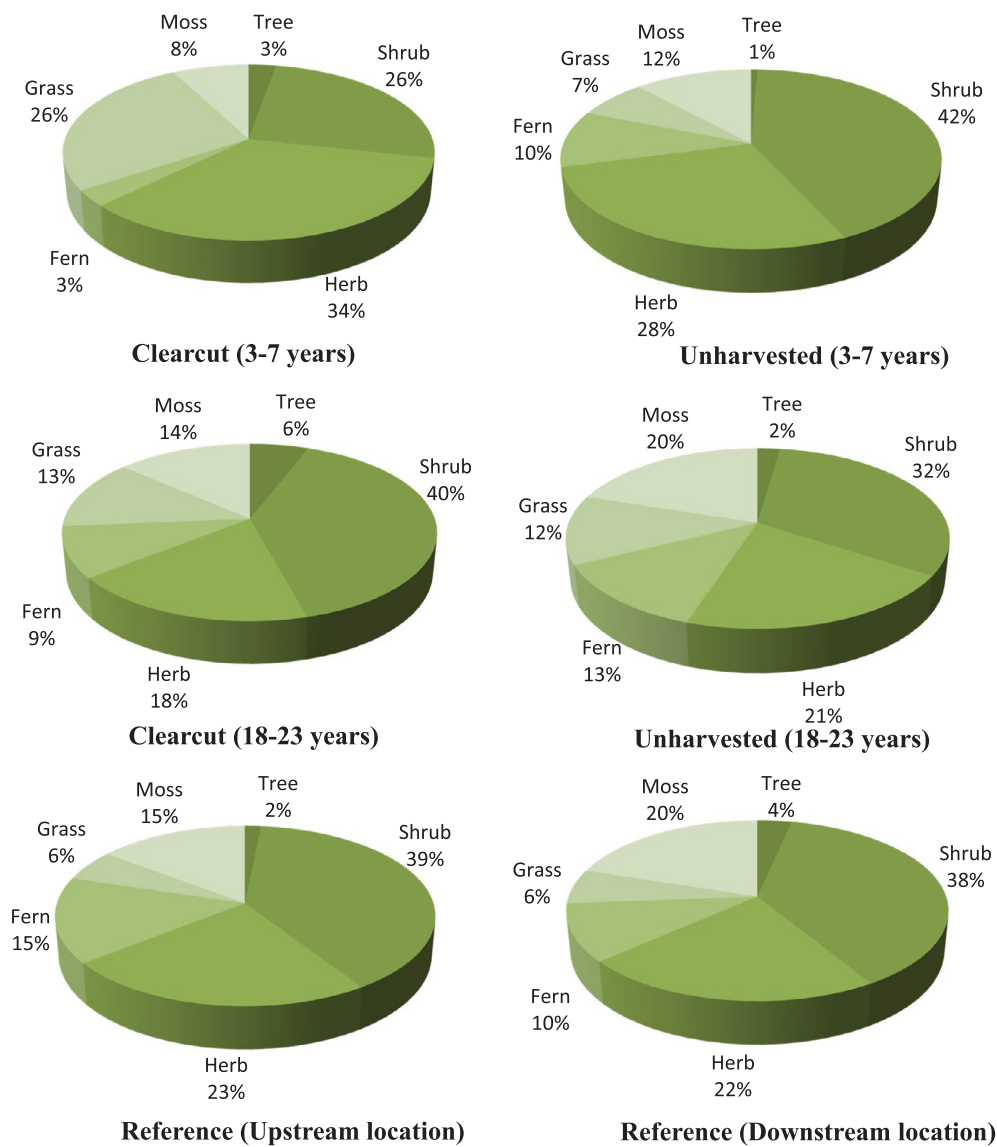
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Appendix 1. Changes in lifeform based species composition (number of species gained or lost) over time since harvesting relative to reference forests of respective locations.

Location		Tree	Shrub	Herb	Fern	Grass	Moss	Total
Reference (Upstream location)	Number of species	4	23	35	13	21	35	131
	Number of species	6	27	63	15	36	33	180
Clearcut (3–7 years)	Species gained	3	9	34	5	16	7	74
		(75.00)	(39.13)	(97.14)	(38.46)	(76.19)	(20.00)	(56.49)
	Species lost	1	5	6	3	1	9	25
		(25.00)	(21.74)	(17.14)	(23.08)	(4.76)	(25.71)	(19.08)
Clearcut (18–23 years)	Number of species	4	32	36	10	26	39	147
	Species gained	1	12	11	2	10	10	46
		(25.00)	(52.17)	(31.43)	(15.38)	(47.62)	(28.57)	(35.11)
	Species lost	1	3	10	5	5	6	30
		(25.00)	(13.04)	(28.57)	(38.46)	(23.81)	(17.14)	(22.90)
Reference (Downstream location)		4	28	35	14	19	37	137
Downstream unharvested (3–7 years)	Number of species	4	37	42	14	26	27	150
	Species gained	1	13	11	2	11	6	44
		(25.00)	(46.43)	(31.43)	(14.29)	(57.89)	(16.22)	(32.12)
	Species lost	1	4	4	2	4	16	31
		(25.00)	(14.29)	(11.43)	(14.29)	(21.05)	(43.24)	(22.63)
Downstream unharvested (18–23 years)	Number of species	5	24	42	13	29	38	151
	Species gained	1	2	15	3	13	8	42
		(25.00)	(7.14)	(42.86)	(21.43)	(68.42)	(21.62)	(30.66)
	Species lost	0	6	8	4	3	7	28
		(0.00)	(21.43)	(22.86)	(28.57)	(15.79)	(18.92)	(20.44)

Note: Figures in parentheses indicate percent of the total number of species recorded in respective reference locations.

Appendix 2. Distribution (mean total cover) of different life forms at early and late clearcut sites along with associated downstream unharvested sites and reference forests of upstream and downstream locations.



Appendix 3. Relative dominance of species at different age classes since clearcut harvesting used in Indicator Species Analysis.

Species	Lifeform	3–7 years old		18–23 years old		Reference forest	
		Clearcut	DS unhar.	Clearcut	DS unhar.	US location	DS location
<i>Abies balsamea</i>	Tree	0.22	0.06	1.26	1.25	0.91	1.22
<i>Betula papyrifera</i>	Tree	0.36	0.00	1.99	0.41	0.00	0.49
<i>Fraxinus nigra</i>	Tree	0.00	0.16	0.00	0.07	0.46	1.04
<i>Picea mariana</i>	Tree	0.14	0.34	0.74	0.27	0.20	0.72
<i>Populus tremuloides</i>	Tree	2.04	0.10	2.25	0.35	0.06	0.00
<i>Acer spicatum</i>	Shrub	0.07	0.52	2.84	3.30	15.87	15.18
<i>Alnus incana</i>	Shrub	1.87	19.08	13.17	10.72	3.78	6.76
<i>Alnus viridis</i>	Shrub	1.45	0.18	0.00	0.02	0.84	0.40
<i>Amelanchier stolonifera</i>	Shrub	0.21	0.99	0.03	0.23	0.42	0.07
<i>Amelanchier spp.</i>	Shrub	0.09	0.00	0.06	0.00	0.00	0.00
<i>Cornus stolonifera</i>	Shrub	2.65	5.32	3.31	3.76	1.19	3.03
<i>Corylus cornuta</i>	Shrub	3.16	3.66	4.98	1.42	7.62	2.97
<i>Diervilla lonicera</i>	Shrub	0.28	1.29	1.87	1.08	0.00	0.21
<i>Ledum groenlandicum</i>	Shrub	0.02	0.13	0.07	0.14	0.22	0.19
<i>Lonicera villosa</i>	Shrub	0.03	0.17	0.18	0.28	0.11	0.02
<i>Prunus pensylvanica</i>	Shrub	1.36	0.35	0.63	0.00	0.18	0.00

<i>Prunus virginiana</i>	Shrub	0.56	0.23	0.09	0.08	0.00	0.00
<i>Rhamnus alnifolia</i>	Shrub	0.00	0.51	0.04	0.50	1.39	1.75
<i>Ribes glandulosum</i>	Shrub	0.04	0.02	0.09	0.15	0.08	0.11
<i>Ribes hirtellum</i>	Shrub	0.59	0.90	0.22	0.60	0.43	0.62
<i>Rosa acicularis</i>	Shrub	0.63	0.76	1.55	0.62	1.53	0.75
<i>Rubus acaulis</i>	Shrub	1.59	0.80	0.10	0.10	2.98	3.35
<i>Rubus idaeus</i>	Shrub	9.34	3.93	6.21	3.62	0.98	0.59
<i>Rubus pubescens</i>	Shrub	4.13	1.80	4.28	5.69	0.49	0.71
<i>Salix spp.</i>	Shrub	0.35	0.72	0.29	0.00	0.54	0.30
<i>Sorbus decora</i>	Shrub	0.05	0.12	0.06	0.05	0.00	0.05
<i>Vaccinium angustifolium</i>	Shrub	0.25	0.17	0.09	0.07	0.00	0.02
<i>Viburnum edule</i>	Shrub	0.34	0.91	0.10	0.50	1.06	1.12
<i>Actaea rubra</i>	Herb	0.06	0.39	0.00	0.04	0.39	0.19
<i>Aralia nudicaulis</i>	Herb	0.10	1.03	0.00	0.18	1.39	0.85
<i>Aster ciliolatus</i>	Herb	0.97	0.12	0.07	0.04	0.17	0.07
<i>Aster macrophyllus</i>	Herb	6.08	3.54	4.75	3.83	1.95	1.65
<i>Aster nemoralis</i>	Herb	0.68	0.17	0.00	0.14	0.00	0.00
<i>Aster umbellatus</i>	Herb	0.01	0.10	0.68	0.50	0.31	0.02
<i>Circaea alpina</i>	Herb	0.00	0.00	0.00	0.00	0.83	1.42
<i>Cirsium muticum</i>	Herb	1.84	0.20	0.33	0.37	0.00	0.00
<i>Clintonia borealis</i>	Herb	0.87	3.25	1.94	1.36	3.23	1.48
<i>Coptis trifolia</i>	Herb	0.36	0.22	0.18	0.20	0.13	0.04
<i>Cornus canadensis</i>	Herb	0.92	0.02	0.38	0.65	0.12	0.01
<i>Epilobium angustifolium</i>	Herb	4.11	0.00	0.18	0.16	0.00	0.00
<i>Fragaria virginiana</i>	Herb	0.30	0.09	0.00	0.00	0.26	0.03
<i>Galium asprellum</i>	Herb	3.35	2.49	1.46	1.27	1.71	1.58
<i>Galium trifidum</i>	Herb	0.35	0.19	0.04	0.18	0.08	0.00
<i>Galium triflorum</i>	Herb	1.88	0.56	1.24	1.42	1.23	1.31
<i>Geum rivale</i>	Herb	0.16	0.10	0.03	0.00	0.10	0.07
<i>Lycopus uniflorus</i>	Herb	0.13	0.14	0.23	0.67	0.17	0.24
<i>Maianthemum canadense</i>	Herb	0.48	1.18	0.19	0.05	0.57	0.85
<i>Mentha arvensis</i>	Herb	0.29	0.20	0.00	1.11	0.28	0.21
<i>Mertensia paniculata</i>	Herb	2.25	2.80	1.69	1.44	1.80	2.57
<i>Mitella nuda</i>	Herb	0.66	2.29	0.56	1.11	1.57	1.85
<i>Petasites frigidus</i>	Herb	0.65	0.80	1.02	1.07	0.32	1.09
<i>Streptopus roseus</i>	Herb	0.61	1.61	0.03	0.06	0.66	1.04
<i>Thalictrum dasycarpum</i>	Herb	1.13	2.85	1.61	2.22	1.03	0.41
<i>Trientalis borealis</i>	Herb	0.19	0.61	0.19	0.04	0.54	0.32
<i>Trillium cernuum</i>	Herb	0.22	0.23	0.02	0.00	0.55	0.23
<i>Viola adunca</i>	Herb	0.49	0.72	0.12	0.60	1.10	1.21
<i>Viola blanda</i>	Herb	0.21	0.88	0.24	0.12	0.92	0.83
<i>Viola macloskeyi</i>	Herb	0.01	0.10	0.03	0.43	0.00	0.05
<i>Viola nephrophylla</i>	Herb	0.26	0.20	0.00	0.03	0.54	0.58
<i>Viola pubescens</i>	Herb	0.30	0.22	0.07	0.05	0.31	0.39
<i>Viola renifolia</i>	Herb	1.06	0.89	0.89	1.42	1.82	1.79
<i>Viola septentrionalis</i>	Herb	0.06	0.02	0.00	0.00	0.00	0.00
<i>Athyrium flexifemina</i>	Fern	1.32	5.66	5.01	7.62	6.14	5.68
<i>Dryopteris carthusiana</i>	Fern	0.08	0.07	0.00	0.04	0.14	0.18
<i>Equisetum pretense</i>	Fern	0.23	0.11	0.00	0.00	1.40	0.83
<i>Equisetum sylvaticum</i>	Fern	0.38	1.74	0.03	0.09	0.34	0.54
<i>Gymnocarpium dryopteris</i>	Fern	0.11	0.26	0.18	0.34	0.20	0.54
<i>Lycopodium annotinum</i>	Fern	0.30	1.20	0.93	1.12	0.51	0.02
<i>Lycopodium dendroideum</i>	Fern	0.12	0.53	0.03	0.03	0.00	0.08
<i>Lycopodium lucidulum</i>	Fern	0.00	0.06	0.30	0.60	0.20	0.48
<i>Matteuccia struthiopteris</i>	Fern	0.29	0.10	0.20	0.51	0.07	0.09
<i>Osmunda claytoniana</i>	Fern	0.00	0.01	2.11	1.16	4.46	0.69
<i>Phlegopteris connectilis</i>	Fern	0.17	0.44	1.02	1.31	1.87	1.28
<i>Carex aenea</i>	Sedge	1.51	0.01	0.69	0.35	0.09	0.49
<i>Carex aquatilis</i>	Sedge	0.91	0.00	0.63	0.65	0.11	0.09
<i>Carex canescens</i>	Sedge	0.32	0.03	0.01	0.00	0.00	0.09
<i>Carex crawfordii</i>	Sedge	0.43	0.50	0.03	0.03	0.26	0.55
<i>Carex crinita</i>	Sedge	0.20	0.26	0.25	0.06	0.00	0.24
<i>Carex deflexa</i>	Sedge	0.66	1.15	0.67	1.44	1.33	1.26
<i>Carex disperma</i>	Sedge	0.35	0.02	0.23	0.04	0.28	0.00
<i>Carex flava</i>	Sedge	0.48	0.68	0.35	0.95	0.17	0.14
<i>Carex gracillima</i>	Sedge	0.28	0.08	0.15	0.08	0.03	0.00
<i>Carex houghtoniana</i>	Sedge	0.24	0.00	0.46	0.43	0.00	0.01
<i>Carex interior</i>	Sedge	0.27	0.06	0.03	0.11	0.00	0.08
<i>Carex intumescens</i>	Sedge	2.02	1.19	2.53	2.97	1.65	1.87
<i>Carex lasiocarpum</i>	Sedge	1.62	0.13	0.14	0.00	0.00	0.00
<i>Carex laxiflora</i>	Sedge	1.03	0.02	0.05	0.20	0.07	0.00
<i>Carex spp.</i>	Sedge	0.16	0.32	0.00	0.05	0.52	0.16
<i>Agrostis scabra</i>	Grass	0.39	0.00	0.28	0.13	0.00	0.00
<i>Bromus ciliatus</i>	Grass	2.06	0.62	0.98	1.30	0.41	0.31
<i>Calamagrostis canadensis</i>	Grass	6.08	0.06	4.51	1.95	0.00	0.00
<i>Cinna latifolia</i>	Grass	0.74	0.00	0.11	0.21	0.02	0.07
<i>Elymus trachycaulus</i>	Grass	0.26	0.04	0.13	0.07	0.06	0.00
<i>Elymus repens</i>	Grass	1.58	0.03	0.47	0.13	0.02	0.32
<i>Glyceria borealis</i>	Grass	0.69	0.00	0.10	0.01	0.02	0.00
<i>Oryzopsis asperifolia</i>	Grass	1.51	0.32	0.00	0.24	0.08	0.07

<i>Graminoide spp.</i>	Grass	2.80	1.54	0.38	0.95	0.56	0.46
<i>Aulacomnium palustre</i>	Moss	0.67	1.10	0.23	0.23	0.12	0.73
<i>Brachythecium reflexum</i>	Moss	0.10	0.07	0.24	0.29	0.01	0.22
<i>Brachythecium rivulare</i>	Moss	0.09	0.15	0.19	0.15	0.10	0.60
<i>Brachythecium turgidum</i>	Moss	0.22	0.00	0.34	0.35	0.03	0.06
<i>Bryum pseudotriquetrum</i>	Moss	1.62	0.04	0.52	1.02	0.13	0.22
<i>Callicladium spp.</i>	Moss	0.17	0.36	0.81	0.53	0.67	0.70
<i>Climacium dendroides</i>	Moss	0.05	0.66	0.00	0.13	0.09	0.15
<i>Dicranum flagellare</i>	Moss	0.04	0.00	0.30	0.26	0.26	0.58
<i>Dicranum fuscescens</i>	Moss	0.29	0.00	0.64	0.34	0.15	0.16
<i>Dicranum polysetum</i>	Moss	0.09	0.22	0.07	0.45	0.22	0.04
<i>Dicranum scoparium</i>	Moss	0.00	0.01	0.32	0.63	0.01	0.00
<i>Dicranum spp.</i>	Moss	0.10	0.14	0.03	0.18	0.13	0.06
<i>Drepanocladus uncinatus</i>	Moss	0.32	0.37	0.30	0.69	0.31	0.25
<i>Eurhynchium pulchellum</i>	Moss	0.82	1.89	2.44	3.40	2.58	5.16
<i>Fissidens spp.</i>	Moss	0.21	0.14	0.24	0.30	0.19	0.88
<i>Hypnum revolutum</i>	Moss	0.00	0.22	0.00	0.37	0.07	0.18
<i>Mnium spp.</i>	Moss	0.14	0.09	0.62	0.64	0.13	0.20
<i>Plagiomnium spp.</i>	Moss	0.65	1.82	0.70	2.04	1.88	2.37
<i>Pleurozium schreberi</i>	Moss	0.46	0.65	1.22	1.31	0.94	0.59
<i>Polytrichum commune</i>	Moss	0.74	0.00	0.02	0.20	0.03	0.00
<i>Polytrichum strictum</i>	Moss	0.00	0.00	0.16	0.00	0.02	0.14
<i>Pylaisiella polyantha</i>	Moss	0.00	0.00	0.08	0.04	0.19	0.40
<i>Rhizomnium pseudopunctatum</i>	Moss	0.16	0.35	0.06	0.41	1.00	0.44
<i>Rhizomnium punctatum</i>	Moss	0.09	0.40	0.07	0.36	0.64	0.38
<i>Rhodobryum roseum</i>	Moss	0.11	0.72	0.37	0.85	0.43	0.52
<i>Rhytidiadelphus triquetrus</i>	Moss	0.20	0.70	0.54	0.85	0.17	0.52
<i>Sphagnum angustifolium</i>	Moss	0.00	0.00	1.27	1.37	0.00	0.00
<i>Sphagnum spp.</i>	Moss	0.09	0.57	0.35	0.63	1.50	2.75
<i>Thuidium delicatulum</i>	Moss	0.13	0.68	0.20	0.75	1.59	1.03

Note: DS = downstream; US = upstream; unhar = unharvested.

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